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SoilStream - A Smart Solar-Powered IoT System for Real-Time Soil Moisture Monitoring and Automated Irrigation

Group No.12

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Project Guide

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Abstract

SoilStream is a smart, solar-powered IoT system designed to address major challenges in agriculture such as inefficient water use, unpredictable weather, and crop diseases. It uses IoT sensors to monitor real-time soil moisture and automate irrigation, reducing water wastage and ensuring optimal crop growth. A machine learning model predicts rainfall probability, helping farmers plan irrigation more efficiently. The system also provides crop-specific soil-water recommendations. through a rule-based approach to enhance productivity. Farmers can upload crop images to a cloud-based AI model, which diagnoses pests and diseases and suggests suitable treatments. By combining IoT, predictive analytics, and AI-driven diagnostics, SoilStream empowers farmers with accurate insights, minimizes resource loss, and improves crop yield and health.

Sustainable Development Goals(SDG) mapped

- **SDG 12 – Responsible Consumption and Production** : Efficient water and resource use, along with precision farming, minimize waste and promote sustainable agricultural.

Introduction

Agriculture faces challenges like inefficient water use, uncertain rainfall, and frequent crop diseases. Current solutions—manual irrigation, basic weather updates, and traditional pest control—are either unreliable or resource-heavy. This motivated us to create SoilStream, a smart solar-powered system combining IoT, Machine Learning, and AI. It monitors soil moisture in real time, automates irrigation to reduce water wastage, predicts rainfall using ML, provides crop-specific soil recommendations via a rule-based system, and diagnoses pests/diseases from farmer-uploaded images through cloud-based AI.

Objectives

- To monitor soil moisture in real-time using IoT based sensors.
- To predict rainfall probability with a machine learning model using weather data, optimizing irrigation to conserve water.
- To provide crop-specific soil-water condition recommendations based on moisture data analysis, enhancing yield using a LLM.
- To diagnose crop pests/diseases via Kindwise API analyzing farmer-uploaded photos, offering tailored treatment suggestions in-app.

Literature Review

Sr no.	Title	Author(s)	Year	Methodology	Drawback
[1]	Smart Irrigation System with IoT, Machine Learning, and Solar Power for Efficient Plant Care	Justin M. A. Capcha-Ochoa et al.	2025	IoT (**ESP32, ESP32-CAM), ML (modified YOLOv3-tiny for plant health detection), solar power, integrated with Telegram + YouTube Live for monitoring**. Achieved 93.86% accuracy, 43.7% water savings.	High complexity and cost (AI + vision + IoT + solar). Requires stable internet for cloud + live monitoring.
[2]	IoT-enabled solar-powered smart irrigation for precision agriculture	Md. Rasel Al Mamun, Abu Kawsar Ahmed, Sidratul Muntaha Upoma, Md. Mashurul Haque, Muhammad Ashik-E-Rabbani	2025	**Raspberry Pi + ESP8266 sensors, MQTT protocol, solar panel + battery, Django** web app for monitoring. Threshold-based pump activation (45–80% soil moisture).	Battery backup limited (2 hrs at night). System efficiency depends on network connectivity and local conditions.
[3]	Development of IoT and AI based Smart Irrigation System	Yunus Emre Kunt	2025	**ESP32 + IoT sensors, AI (LSTM model for prediction), Blynk app control, integrated with Google Sheets** for data logging.	Computational cost of AI model; initial testing done on prototype, not at full agricultural scale.
[4]	Smart Irrigation System Using IoT and Solar	Talha Aman Ali, Mohd Umer Ahmed Khan, Anas Ahmed, Pradeep Kumar	2025	**ESP32 + Soil Moisture + DHT22, solar-powered pump controlled via Blynk** app. Real-time monitoring and alerts.	Tested only in small-scale setup; system performance under varying weather conditions not studied.

[5]	A solar-powered, internet of things (IoT)-controlled water irrigation system supported by rainfall forecasts utilizing aerosols: a review	D. Balamurali, Samarth Chakankar, Giridhar Sharma, Ananjan Praveen Pagey, M. Natarajan, Saboor Shaik, Sudarshan Gnanavendan, Müslüm Arıcı	2025	Review of IoT-based irrigation systems integrating **solar PV, pollutant-concentration-based rainfall prediction, MPPT controllers, sensors, and reversible pumps**. Focused on sustainable water and energy management, with novel rainfall prediction based on pollutants instead of aerosol size.	High installation cost for small farmers; IoT security concerns; prediction accuracy depends on pollutant data availability.
[6]	Intelligent and automatic irrigation system based on internet of things using fuzzy control technology	Xinying Liu, Zhihuan Zhao, Amin Rezaeipanah	2025	Proposed **IoT-based intelligent irrigation system** with **fuzzy rule-based inference, DNN for decision-making, and fuzzy-based routing (OSPF-enhanced)**. Evaluated via simulations against DLQR, SPIS, FWIS. Showed better network lifetime, energy efficiency, and adaptability.	Higher computational demand (DNN training); sensor dependency; complex deployment in real-world farming.
[7]	Smart irrigation system using IoT and machine learning methods	Ahmed A. Esmail, Manar A. Ibrahim, Sayed M. Abdallah, Ahmed E. Radwan, Haya K. Elsonbaty, Mohamed A. Elsayed, Noure A. Elnakeib, Merit Samir Dawoud, Amir El-Ghamry, Khaled M. Fouad, Ibrahim F. Moawad	2023	Built a **smart irrigation system using IoT sensors + ML (decision tree, other models)**. Dataset: “Crop Irrigation Scheduling” from Kaggle with attributes (Crop type, Crop days, Soil moisture, Temperature, Humidity, Irrigation). Integrated IoT for real-time monitoring + API for management.	ML model performance depends on dataset quality; scalability to larger fields untested; decision tree better than others but may not generalize across all crops.

[8]	Smart irrigation system based on IoT and machine learning	Youness Tace, Mohamed Tabaa, Sanaa Elfilali, Cherkaoui Leghris, Hassna Bensag, Eric Renaulte	2022	IoT-based sensors (humidity, temperature, rain) **with ML models (KNN, Logistic Regression, Neural Networks, SVM, Naïve Bayes). KNN achieved 98.3% accuracy.** Web app for monitoring.	Model tested in controlled environment; scalability and adaptability to diverse climates not validated.
[9]	Design and implementation of solar-powered with IoT-Enabled portable irrigation system	Roshahliza M. Ramli, Waheb A. Jabbar	2022	Developed solar-powered portable pump with **NodeMCU, soil/moisture sensors, IoT-enabled** monitoring via Blynk app.	Limited power during low sunlight; portability suitable for small farms but may not scale well to large farms.
[10]	IoT based solar powered smart irrigation system	P.K. Devan, Arun K, Arvindkumar N.H, Aravind R, Dinesh Kumar R	2021	PIC microcontroller + **ESP8266 Wi-Fi module, solar-powered water pump,** soil moisture sensors.	Limited scalability; designed as proof-of-concept model.
[11]	Implementation of artificial intelligence in agriculture for optimisation of irrigation and application of pesticides and herbicides	Tanha Talaviya, Dhara Shah, Nivedita Patel, Hiteshri Yagnik, Manan Shah	2020	Survey of **AI applications in agriculture (robots, drones, image recognition, soil water sensing, automated weeding**, irrigation techniques).	Limited real-world deployment due to high cost, need for skilled operators, and scalability issues.

[12]	Solar System Powered IoT Solution for Smart Irrigation	Ritesh Maskara, Nikhil Maskara, Shuvabrata Bandopadhaya	2019	IoT + Solar + Arduino Mega 2560, GSM module, soil moisture sensors. Prototype system tested.	Farmers may not track conditions in real time; limited adoption due to infrastructure and connectivity issues.
[13]	Smart Irrigation Control System Using Internet of Things	Soroor Al-Zahrani; Heyam H. Al-Baity	2019	IoT-based smart irrigation using sensors to monitor soil and automate irrigation.	Limited to small-scale study; lacks integration with predictive analytics and renewable power sources.
[14]	Solar Powered Smart Irrigation System Using Internet of Things	Kshirod Rout, Sivkumar Mishra, et al.	2018	IoT sensors integrated with solar-powered pumps. Demonstrated in prototype with soil moisture-based irrigation control.	Early work – lacks AI/ML integration and advanced data analytics.
[15]	Intelligent Irrigation System Based on Arduino	Carlos G. Casado; Máximo L. Cisquella; Sergio A. López	2018	Arduino-based soil moisture sensing and irrigation control system. Low-cost prototype.	Limited scalability; no wireless IoT integration; not energy-efficient

[16]	IOT based Smart Irrigation System	Srishti Rawal	2017	Arduino + IoT sensors (soil moisture, temperature) with cloud-based control via mobile app.	Lacked machine learning/weather prediction; energy dependency on external power.
[17]	IoT based Crop-Field Monitoring and Irrigation Automation	P. Rajalakshmi; S. Devi Mahalakshmi	2016	Wireless sensor network (soil, humidity, temperature) integrated with IoT for real-time monitoring and irrigation automation.	Prototype-level system; not tested for long-term field deployment; no solar integration .

Research Gap

Survey of Current Solutions:

- Most systems only react to dry soil; they don't proactively check for predicted rainfall, leading to water waste.Most systems only react to dry soil, leading to water waste.Often rely only on soil moisture, ignoring key atmospheric data.
- Commercial systems are expensive and closed. DIY projects often lack user-friendly apps

Is the Problem Solved?

No. Basic automation is solved, but the core challenge of an intelligent, predictive, affordable, and integrated crop management system remains largely unaddressed.

Problem Definition

The goal of the SoilStream project is to develop an intelligent, solar-powered IoT agricultural system that integrates real-time soil moisture monitoring, machine learning-based rainfall prediction, automated irrigation control, and AI-driven crop disease diagnosis to address critical challenges in modern agriculture including water wastage from inefficient irrigation practices, unpredictable weather patterns disrupting crop planning, and delayed pest/disease identification leading to significant yield losses while providing farmers with data-driven insights for sustainable and productive farming.

Key points to address:

- Real-Time Soil Moisture Monitoring using IoT Sensors
- ML-Based Rainfall Prediction for Irrigation Optimization
- Automated Solar-Powered Irrigation Control System
- AI-Powered Crop Disease & Pest Diagnosis via Image Analysis
- Cross-Platform Mobile Application for Farm Management

Scope

1. SoilStream can be applied in various agricultural environments such as small-scale farms, organic cultivation, and research plots, providing comprehensive precision irrigation and crop monitoring solutions.
2. SoilStream caters to small-scale farmers, agricultural students, and organic farming enthusiasts, offering tailored soil monitoring and irrigation automation that adapts to different expertise levels and crop types.
3. With real-time monitoring, predictive analytics, and AI diagnostics, SoilStream reduces water wastage and makes precision agriculture accessible for farmers seeking sustainable yield optimization.

Technological Stack

1. On-Field Hardware Unit:-

- **Microcontroller:** ESP32-WROOM-32

Sensors:

- **Capacitive Soil Moisture Sensor:** For accurate, corrosion-resistant soil moisture readings.
- **BME280:** For atmospheric data (temperature, humidity, barometric pressure).

Actuator: 5V Relay Module & Submersible Water Pump(5-12V).

Power System: Solar Panel (6V,5W), TP4056 Charge Controller, 18650 Li-ion Battery, and a 5V Boost Converter.

2. Firmware Development:-

The software running on the ESP32 to control its operations.

- **Programming Language:** C++, Python(For building ML model)
- **Development Environment:** ArduinoIDE.
- **Key Libraries:**
  - **Firebase-ESP-Client:** To handle the real-time communication with the Firebase backend.
  - **Adafruit_BME280:** To interface with the atmospheric sensor.

3. Cloud Backend & Services:-

The serverless, scalable backbone of the project.

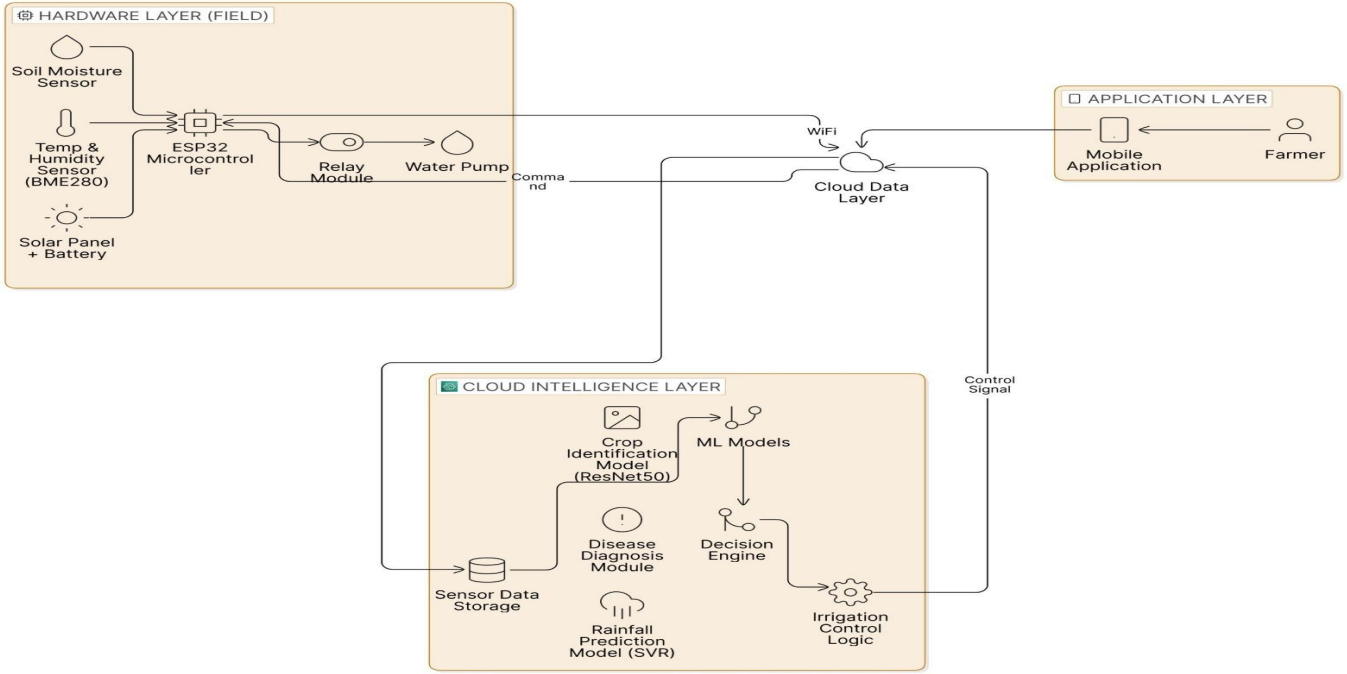
- **Platform:** Google Firebase
- **Database:** Firestore/ Realtime Database
- **Serverless Logic:** Firebase Cloud Functions for executing the ML model and rule-based logic.

4. Mobile Application (Frontend):-

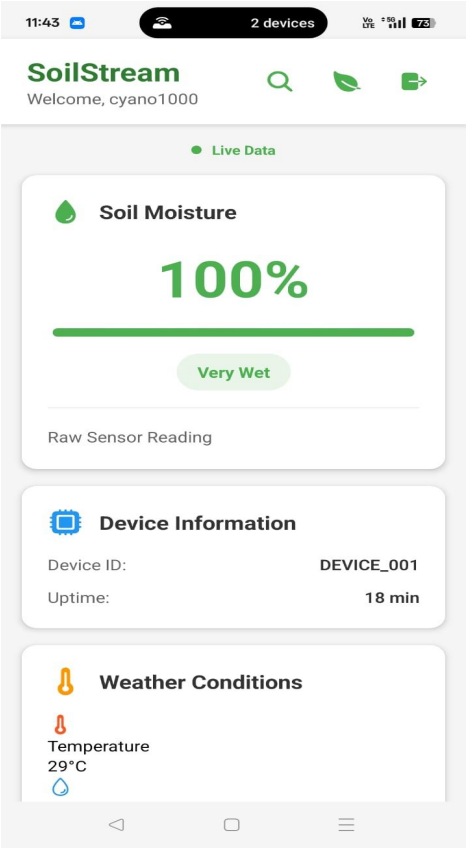
The cross-platform user interface for monitoring and control.

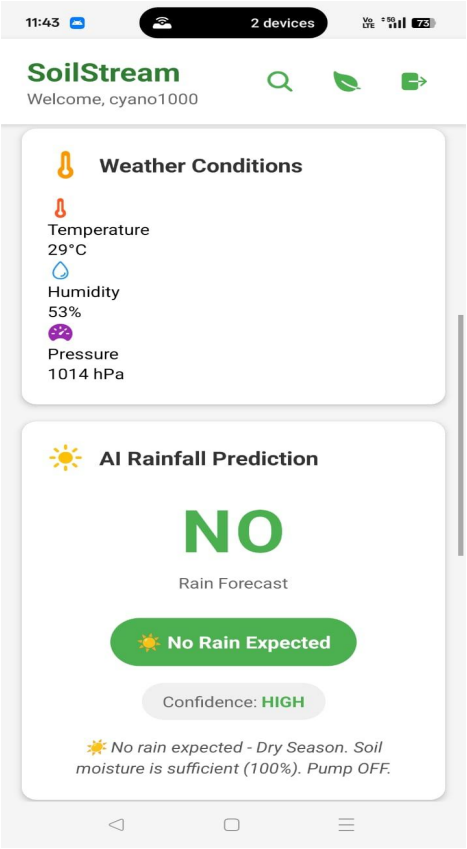
- **Framework:** React Native
- **Programming Language:** JavaScript

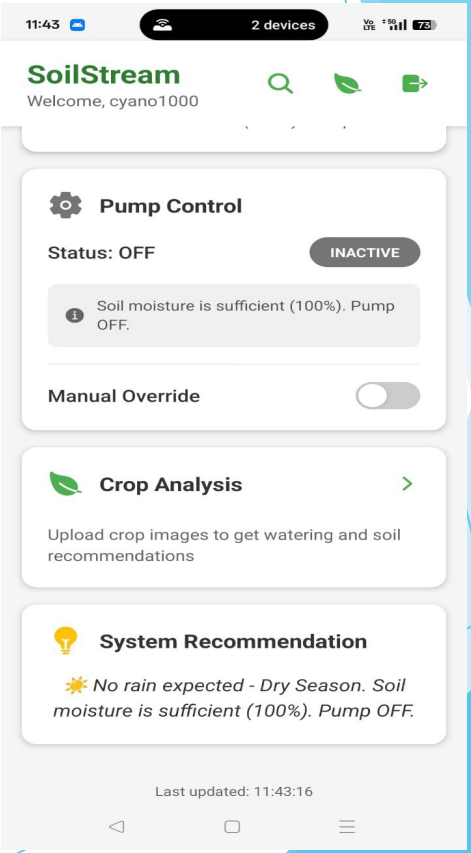
Proposed system architecture

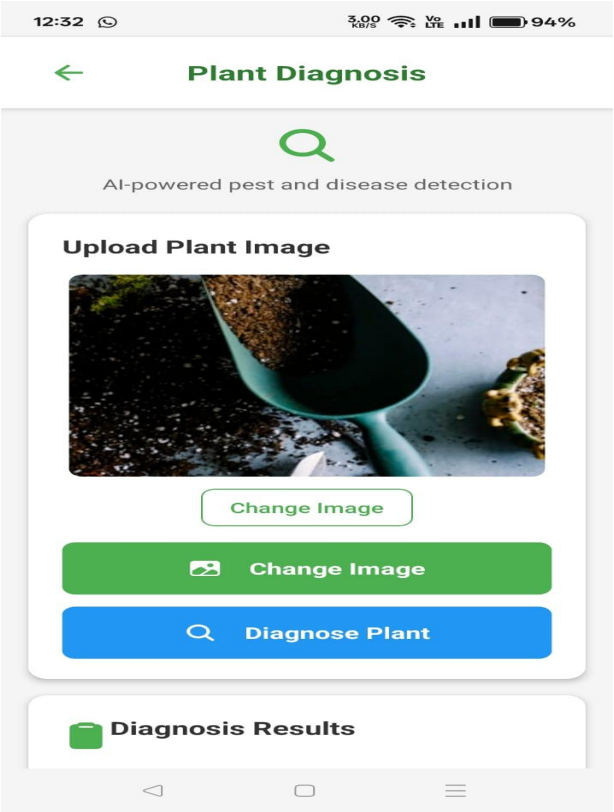


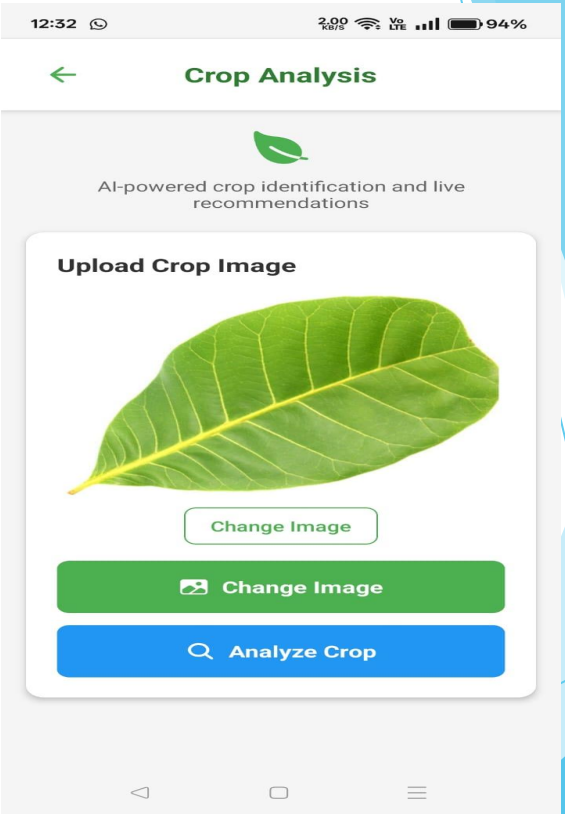
Implementation








12:32
94%
Plant Diagnosis
AI-powered pest and disease detection
Upload Plant Image
Change Image
Change Image
Diagnose Plant
Diagnosis Results


12:32
94%
Crop Analysis
AI-powered crop identification and live recommendations
Upload Crop Image
Change Image
Change Image
Analyze Crop

Review Suggestions

1) Propose every crop in the region of Maharashtra we are focusing.

2) Suggest remedies for crop disease prediction.

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Thank You...!!